Lab Session 02

Practice any project management tool to prepare a project plan

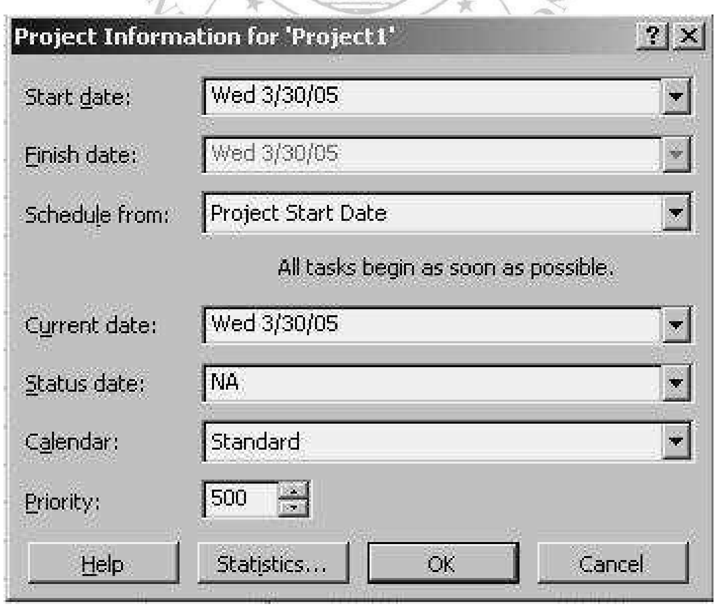
Project management

Project management is the process of planning, organizing, and managing tasks and resources to accomplish a defined objective, usually within constraints on time, resources, or cost. Sound planning is one key to project success, but sticking to the plan and keeping the project on track is no less critical. Microsoft Project features can help in monitoring progress and keep small slips from turning into landslides.

First let us understand some basic terms and definitions that will be used quite often:

Scope: of any project is a combination of all individual tasks and their goals.
Resources: can be people, equipment, materials or services that are needed to complete various tasks. The amount of resources affects the scope and time of any project.

You can create a Project from a Template File by choosing File > New fromthe menu. In the New File dialog box that opens, select the Project Templates tab and select the template that suits your project best and click OK. (You may choose a Blank Project Template and customize it)
Once a new Project page is opened, the Project Information dialog box opens.



***Tasks*:** They are a division of all the work that needs to be completed in order to accomplish the project goals.

STARTING A NEW PROJECT

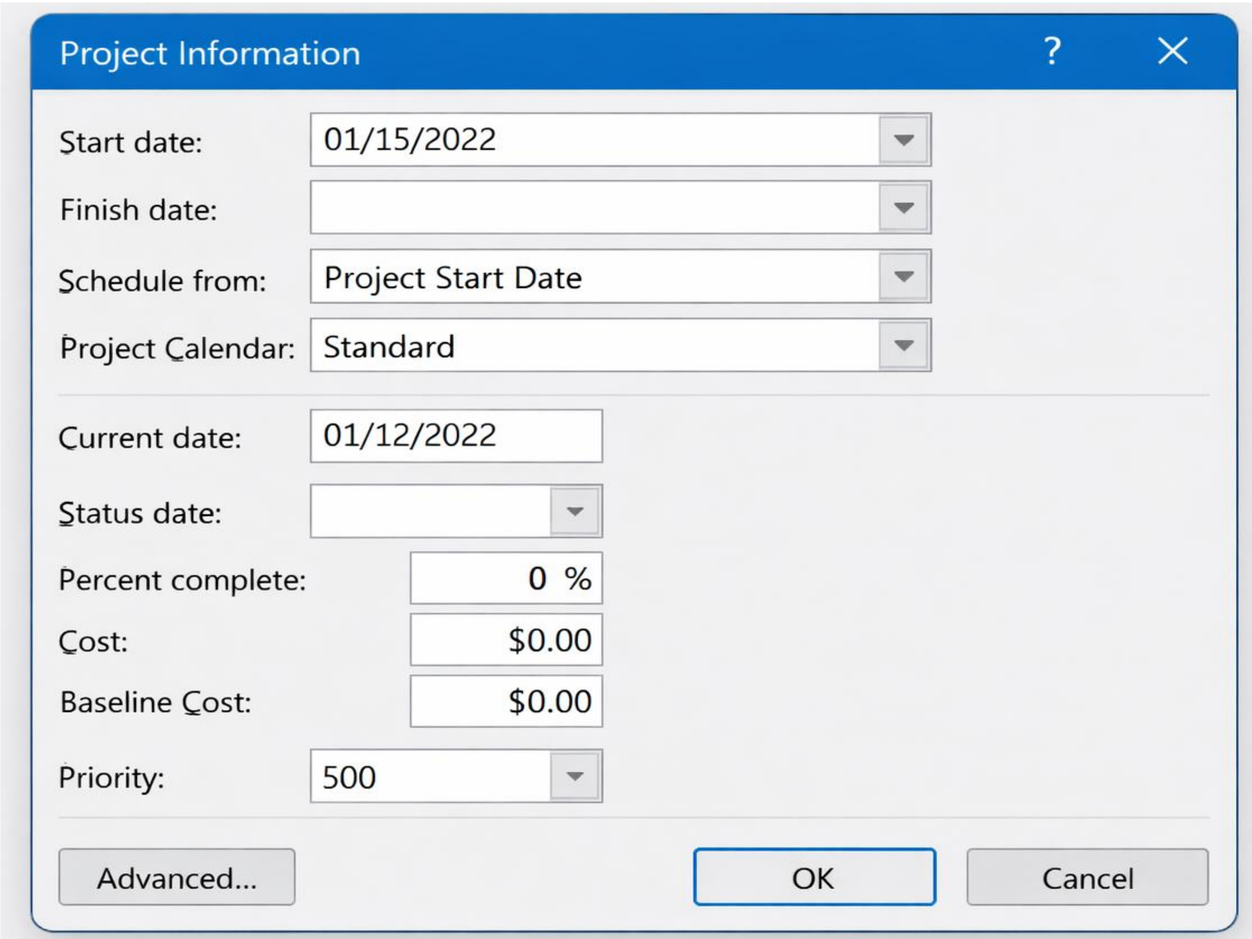


Fig 2.1: Project Information dialog box

Enter the start date or select an appropriate date by scrolling down the list. Click OK. Project automatically enters a default start time and stores it as part of the dates entered and the application window is displayed.

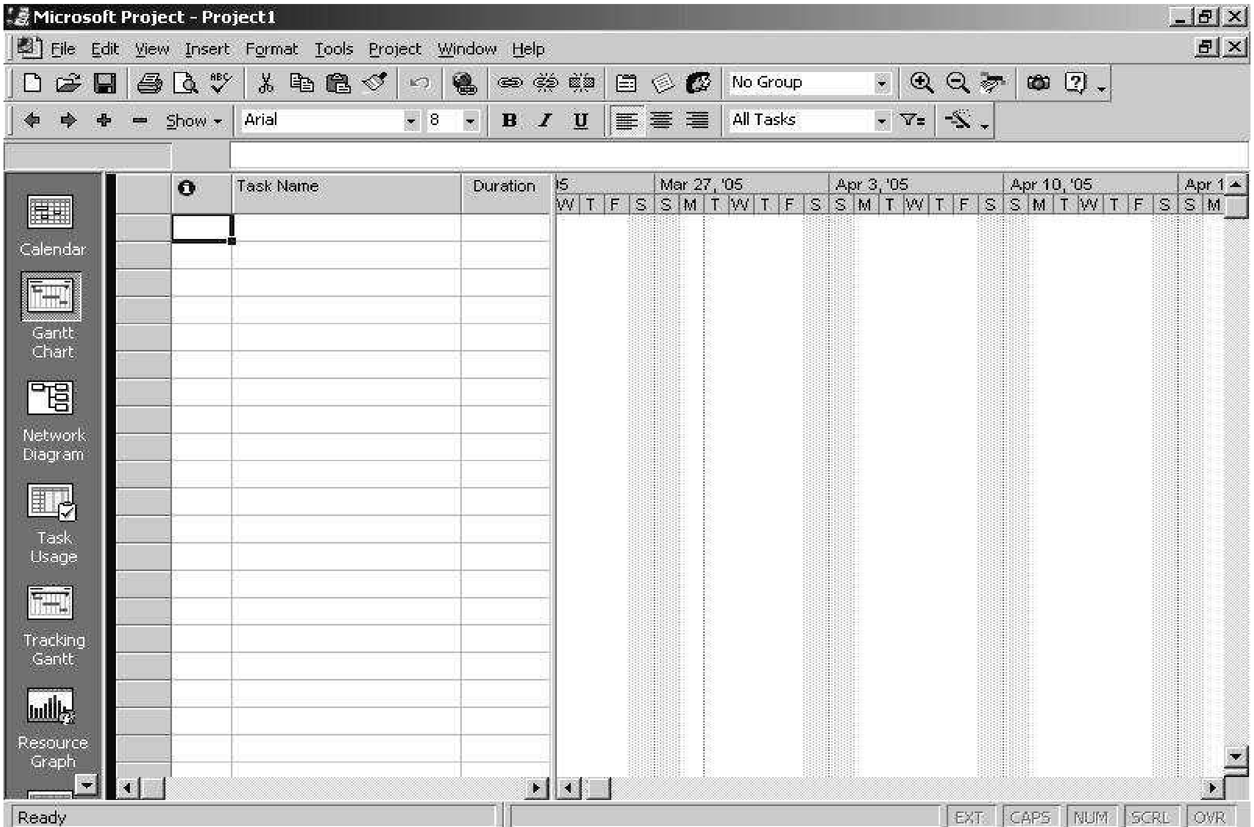


Fig 2.2: Main Window

Views

Views allow you to examine your project from different angles based on what information you want displayed at any given time. You can use a combination of views in the same window at the same time.

Project Views are categorized into two types:

- Task Views (5 types)
- Resource Views (3 types)

The Project worksheet is located in the main part of the application window and displays different information depending on the view you choose. The default view is the Gantt Chart View.

Tasks

Goals of any project need to be defined in terms of tasks. There are four major types of tasks:

1. *Summary tasks* - contain subtasks and their related properties
2. *Subtasks* - are smaller tasks that are a part of a summary task
3. *Recurring tasks* - are tasks that occur at regular intervals
4. *Milestones* - are tasks that are set to zero duration and are like interim goals in the project

- ***Entering Tasks and assigning task duration***

Click in the first cell and type the task name. Press enter to move to the next row. By default, estimated duration of each task is a day. But, there are very few tasks that can be completed in a day's time. You can enter the task duration as and when you decide upon a suitable estimate.

Double-clicking a task or clicking on the Task Information button on the standard toolbar opens the *Task Information box*. You can fill in more detailed descriptions of tasks and attach documents related to the task in the options available in this box.

To enter a *milestone*, enter the name and set its duration to zero. Project represents it as a diamond shape instead of a bar in the Gantt chart.

To copy tasks and their contents, click on the task ID number at the left of the task and copy and paste as usual.

You can enter *Recurring tasks* by clicking on Insert > Recurring task and filling in the duration and recurrence pattern for the task.

Any action you perform on a summary task - deleting it, moving or copying it apply to its subtasks too.

- ***Outlining tasks***

Once the summary tasks have been entered in a task table, you will need to insert subtasks in the blank rows and indent them under the summary task. This is accomplished with the help of the outlining tool.

Outlining is already active when you launch a project and its tools are found at the left end of the Formatting bar. To enter a subtask, enter the task in a blank cell in the Task Name column and click the Indent button on the Outlining tool bar. The Show feature in this toolbar is drop down tool that gives you an option of different Outline levels.

A summary task is outdented to the left cell border, is bold and has a Collapse (-) (Hide subtasks) button in front of it and its respective subtasks are indented with respect to it.

Advantages of Outlining:

- It creates multiple levels of subtasks that roll up into a summary task
- Collapse and expand summary tasks when necessary

- Apply a Work Breakdown structure
- Move, copy or delete entire groups of tasks

LINKS

Tasks are usually scheduled to start as soon as possible i.e. the first working day after the project start date.

Dependencies

Dependencies specify the manner in which two tasks are linked. Because Microsoft Project must maintain the manner in which tasks are linked, dependencies can affect the way a task is scheduled. In a scenario where there are two tasks, the following dependencies exist:

Finish to Start – Task 2 cannot start until task 1 finishes.
Start to Finish – Task 2 cannot finish until task 1 starts.
Start to Start – Task 2 cannot start until task 1 starts.
Finish to Finish – Task 2 cannot finish until task 1 finishes.

Tasks can be linked by following these steps:

1. Double-click a task and open the Task Information dialog box.
2. Click the predecessor tab.
3. Select the required predecessor from the drop down menu.
4. Select the type of dependency from drop down menu in the Type column.
5. Click OK.

The Split task button splits tasks that may be completed in parts at different times with breaks in their duration times.

CONSTRAINTS

Certain tasks need to be completed within a certain date. Intermediate deadlines may need to be specified. By assigning constraints to a task you can account for scheduling problems. There are about 8 types of constraints and they come under the flexible or inflexible category. They are:

- **As Late As Possible** – Sets the start date of your task as late in the Project as possible, without pushing out the Project finish date.
- **As Soon As Possible** – Sets the start date of your task as soon as possible without preceding the project start date.
- **Must Finish On** – Sets the finish date of your task to the specified date.
- **Must Start On** – Sets the start date of your task to the specified date.
- **Start No Earlier Than** – Sets the start date of your task to the specified date or later.
- **Start No Later Than** – Sets the start date of your task to the specified date or earlier.
- **Finish No Earlier Than** – Sets the finish date of your task to the specified date or later.
- **Finish No Later Than** – Sets the finish date of your task to the specified date or earlier.

Flexible constraints (demarcated by a red dot in Microsoft Project 2000) restrict scheduling to a great extent whereas flexible constraints (blue dot) allow Project to calculate the schedule and make appropriate adjustments based on the constraint applied.

Inflexible constraints can cause conflicts between successive and preceding tasks at times and you may need to remove such a constraint.

***To apply a constraint*:**

1. Open the Task Information dialog box.
2. Click the Advanced tab and open the Constraint type list by clicking on the drop-down arrow and select it.
3. Select a date for the Constraint and click OK.

UPDATE COMPLETED TASKS

If you have tasks in your project that have been completed as they were scheduled, you can quickly update them to 100% complete all at once, up to a date you specify.

1. On the View menu, click Gantt Chart.
2. On the Tools menu, point to Tracking, and then click Update Project
3. Click Update work as complete through and then type or select the date through which you want progress updated.
4. If you don't specify a date, Microsoft Project uses the current or status date.
5. Click Set 0% or 100% complete only.
6. Click Entire Project

RESOURCES

Once you determine that you need to include resources into your project you will need to answer the following questions:

- What kind of resources do you need?
- How many of each resource do you need?
- Where will you get these resources?
- How do you determine what your project is going to cost?

Resources are of two types - *work* resources and *material* resources.

Work resources complete tasks by expending time on them. They are usually people and equipment that have been assigned to work on the project.

Material resources are supplies and stocks that are needed to complete a project.
A new feature in Microsoft Project 2000 is that it allows you to track material resources and assign them to tasks.

Entering Resource Information in Project

The Assign Resources dialog box is used to create list of names in the resource pool.

To enter resource lists:

1. Click the Assign Resources button on the Standard Tool bar or Tools > Resources > Assign resources.
2. Select a cell in the name column and type a response name. Press Enter
3. Repeat step 2 until you enter all the resource names.
4. Click OK.

Resource names cannot contain slash (/), brackets [ ] and commas (,). Another way of defining your resource list is through the Resource Sheet View.

1. Display Resource Sheet View by choosing View > Resource Sheet or click the Resource Sheet icon on the View bar.
2. Enter your information. (To enter material resource use the Material Label field)

To assign a resource to a task:

1. Open the Gantt Chart View and the Assign Resources Dialog box.
2. In the Entry Table select the tasks for which you want to assign resources.
3. In the Assign Resources Dialog box, select the resource (resources) you want to assign.
4. Click the Assign button.

EXERCISE

1. Construct a project plan for a project who's SRS is documented in Lab session 1 (Exercise 1). Also attach its printout.

2. Construct a project plan for a project who's SRS is documented in Lab session 1 (Exercise 2). Also attach its printout.